Shrinkage test on faces: a real donor faded towards the dataset mean, blend = mean + lambda (donor - mean), lambda = r of the rung (CelebA: repeat 0.35, mirror 0.51; IMDb-WIKI: repeat 0.30, mirror 0.45). R: 0 = the target itself, 1 = unrelated person.

dataset	metric	another photo R donor	R blend	blend - donor [95% CI]	verdict	own mirror half R donor	R blend	blend - donor [95% CI]	verdict
CelebA	MSE	0.64	0.43	-0.21 [-0.24, -0.18]	prefers the faded donor	0.48	0.36	-0.11 [-0.15, -0.08]	prefers the faded donor
CelebA	RMSE	0.78	0.65	-0.13 [-0.15, -0.12]	prefers the faded donor	0.64	0.58	-0.06 [-0.08, -0.04]	prefers the faded donor
CelebA	MAE	0.78	0.65	-0.13 [-0.14, -0.11]	prefers the faded donor	0.63	0.58	-0.05 [-0.07, -0.03]	prefers the faded donor
CelebA	PSNR	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CelebA	PSNR, fixed range	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
CelebA	SSIM	0.83	0.78	-0.05 [-0.08, -0.02]	prefers the faded donor	0.69	0.70	+0.00 [-0.01, +0.02]	indifferent
CelebA	Pearson r	0.63	0.52	-0.11 [-0.13, -0.08]	prefers the faded donor	0.40	0.37	-0.03 [-0.04, -0.01]	prefers the faded donor
CelebA	ACC vs dataset mean	0.64	0.64	-0.00 [-0.00, +0.00]	indifferent	0.40	0.40	-0.00 [-0.00, +0.00]	indifferent
CelebA	ACC vs sex-age mean	0.66	0.69	+0.03 [+0.02, +0.04]	penalises fading	0.42	0.43	+0.01 [+0.01, +0.02]	penalises fading
CelebA	lDDT-like	0.80	0.62	-0.18 [-0.20, -0.16]	prefers the faded donor	0.67	0.57	-0.11 [-0.13, -0.09]	prefers the faded donor
CelebA	TM-like	0.79	0.67	-0.12 [-0.13, -0.10]	prefers the faded donor	0.62	0.59	-0.03 [-0.05, -0.02]	prefers the faded donor
CelebA	GDT-like	0.76	0.62	-0.14 [-0.16, -0.12]	prefers the faded donor	0.57	0.53	-0.05 [-0.07, -0.02]	prefers the faded donor
CelebA	lDDT, repeat tolerance	0.75	0.53	-0.22 [-0.24, -0.20]	prefers the faded donor	0.58	0.46	-0.12 [-0.15, -0.10]	prefers the faded donor
CelebA	RMSE / repeat tolerance	0.78	0.65	-0.13 [-0.15, -0.12]	prefers the faded donor	0.63	0.58	-0.06 [-0.08, -0.04]	prefers the faded donor
IMDb-WIKI	MSE	0.69	0.45	-0.24 [-0.27, -0.22]	prefers the faded donor	0.53	0.39	-0.14 [-0.19, -0.10]	prefers the faded donor
IMDb-WIKI	RMSE	0.81	0.66	-0.15 [-0.16, -0.13]	prefers the faded donor	0.67	0.60	-0.07 [-0.09, -0.05]	prefers the faded donor
IMDb-WIKI	MAE	0.80	0.66	-0.14 [-0.16, -0.12]	prefers the faded donor	0.67	0.61	-0.06 [-0.09, -0.04]	prefers the faded donor
IMDb-WIKI	PSNR	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
IMDb-WIKI	PSNR, fixed range	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
IMDb-WIKI	SSIM	0.86	0.81	-0.05 [-0.08, -0.02]	prefers the faded donor	0.74	0.74	+0.00 [-0.01, +0.02]	indifferent
IMDb-WIKI	Pearson r	0.68	0.56	-0.12 [-0.15, -0.10]	prefers the faded donor	0.44	0.41	-0.03 [-0.05, -0.02]	prefers the faded donor
IMDb-WIKI	ACC vs dataset mean	0.68	0.68	-0.00 [-0.00, +0.00]	indifferent	0.45	0.45	-0.00 [-0.00, +0.00]	indifferent
IMDb-WIKI	ACC vs sex-age mean	0.71	0.73	+0.02 [+0.01, +0.03]	penalises fading	0.46	0.48	+0.02 [+0.01, +0.03]	penalises fading
IMDb-WIKI	lDDT-like	0.86	0.64	-0.22 [-0.24, -0.20]	prefers the faded donor	0.73	0.60	-0.13 [-0.15, -0.11]	prefers the faded donor
IMDb-WIKI	TM-like	0.81	0.68	-0.13 [-0.15, -0.11]	prefers the faded donor	0.66	0.61	-0.05 [-0.07, -0.02]	prefers the faded donor
IMDb-WIKI	GDT-like	0.78	0.63	-0.15 [-0.17, -0.13]	prefers the faded donor	0.61	0.55	-0.06 [-0.08, -0.03]	prefers the faded donor
IMDb-WIKI	lDDT, repeat tolerance	0.82	0.54	-0.28 [-0.30, -0.25]	prefers the faded donor	0.65	0.49	-0.16 [-0.18, -0.13]	prefers the faded donor
IMDb-WIKI	RMSE / repeat tolerance	0.81	0.66	-0.15 [-0.16, -0.13]	prefers the faded donor	0.67	0.60	-0.07 [-0.09, -0.05]	prefers the faded donor

Intervals: bootstrap over families and window positions. face frame _v6f: visual QC approved by Kate, 6 Nov 2026 (Amendment 10); landmark affine to the population template (face protocol, Limitations 3)